

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                        | Marks available |          |          |           |          |           |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |       |  |                                                                                                                                                                                                                                                                                        | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
| 4.       | (a) | (i)   |  | Suitable scale starting at origin on both axes i.e. plotted points must occupy at least half of the graph paper (1)<br>All points plotted correctly to within <1 small square division (1)<br>Curve of best fit for <u>all</u> points <1 small square division from plotted points (1) | 1               | 1<br>1   |          | 3         | 3        | 3         |
|          |     | (ii)  |  | Number of dice remaining = 33 (taken from candidate's graph) (1)<br>So number removed = 4 (1)<br>Award 1 mark for an answer of 17                                                                                                                                                      |                 | 2        |          | 2         | 1        | 2         |
|          |     | (iii) |  | Line evident on graph at 25 (1)<br>Number of throws read from candidate's graph (1)<br>Accept answer rounded to nearest throw                                                                                                                                                          |                 | 2        |          | 2         | 1        | 2         |
|          |     | (iv)  |  | Evidence of halving from 1 000 <b>or</b> 3 half-lives (1)<br>5.1 ( <b>ecf</b> ) $\times 3 = 15.3$ (1)                                                                                                                                                                                  |                 | 2        |          | 2         | 2        | 2         |
|          | (b) |       |  | <b>Any 2 <math>\times</math> (1) from:</b><br>- decay is random / throwing a dice is random<br>- smooths out fluctuations in data<br>- more accurate results / increase confidence in results<br><br>N.B. Treat reference to anomalies as neutral                                      |                 |          | 2        | 2         |          | 2         |
|          |     |       |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                | <b>1</b>        | <b>8</b> | <b>2</b> | <b>11</b> | <b>7</b> | <b>11</b> |